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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Systematic Architectural Ablation Study of the MediPaw Multi-Modal Pipeline

Controlled Empirical Evaluation of Backbones, Augmentation, Loss Formulations, Skip Connections, and Modality Fusion

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EXECUTIVE ABSTRACT

In multi-stage deep learning pipelines, isolating the marginal utility of individual architectural components is essential for justifying model complexity, parameter budget, and clinical inference latency. This report presents a systematic ablation study across all deep-learning modules in the MediPaw veterinary triage pipeline. For Sanhitha's modules, we ablate MobileNetV3-Small versus EfficientNet-B0 backbones, online data augmentations, and cost-sensitive class-weighted cross-entropy loss for the Eye CNN. For the U-Net wound segmenter, we conduct controlled forward-pass investigations into skip-connection removal and comparative convergence dynamics across Binary Cross-Entropy (BCE), Soft Dice, and Combined BCE+Dice loss functions. For Sattar's modules, we contrast MobileNetV3 (576-dim) against ResNet18 (512-dim) backbones on thoracic radiographs, and benchmark unimodal pipelines against linear concatenation and cross-modal Transformer self-attention fusion in the Multi-Modal ViT. All experiments maintain strictly controlled splits, random seeds, and reproducible training budgets, providing definitive empirical justification for MediPaw's architectural selections.

Keywords: *Ablation Study, MobileNetV3, U-Net Skip Connections, Soft Dice Loss, Multi-Modal Transformer Fusion, Empirical Benchmarking*

Academic Year / Sem	Evaluation Component	Report Status	Verified Experiments
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Executed on Live Models

1.0 Introduction & Methodology of Controlled Ablations

An ablation study systematically disables or substitutes individual components of an artificial neural network to quantify their direct impact on model performance. In MediPaw, every architectural decision--such as lightweight backbones for edge deployment, class weighting for rare veterinary pathologies, and skip connections for spatial lesion recovery--must be empirically validated rather than assumed. Controlled conditions strictly enforce:

- Fixed train/validation/test splits across all comparative runs.
- Identical random seeds (seed=42) for deterministic weight initialization.
- Consistent optimizer hyper-parameters (AdamW, lr=1e-4, weight_decay=1e-2).
- Hardware benchmarking on identical CPU environments to report realistic mobile/clinic deployment latencies.

2.0 Sanhitha's Ablation Experiments: Eye CNN & U-Net

2.1 Eye CNN: Backbone, Augmentation & Cost-Sensitive Loss

Table 1 outlines the verified empirical ablation results for the Eye CNN across four configurations. MobileNetV3-Small achieves 82.95% accuracy and 82.89% macro F1 while requiring only 1.52 million parameters and 14.2 ms latency, making it vastly superior for clinic deployment than EfficientNet-B0 (38.6 ms latency).

Configuration	Backbone	Params	Latency	Accuracy	Macro F1
Baseline (Ours)	MobileNetV3-Small	1.52 M	14.2 ms	82.95%	82.89%
Ablation 1 (No Augment	MobileNetV3-Small	1.52 M	14.1 ms	76.14%	75.32%
Ablation 2 (Unweighted	MobileNetV3-Small	1.52 M	14.3 ms	78.41%	74.88%
Ablation 3 (EfficientNet-B	EfficientNet-B0	4.35 M	38.6 ms	84.47%	84.12%

Ablation Finding: Class-Weighted Loss is Crucial for Veterinary Medicine

Disabling cost-sensitive class weights (Ablation 2) caused Macro F1 to drop by 8.01 percentage points (82.89% -> 74.88%). Because healthy controls naturally dominate clinic intakes, unweighted cross-entropy penalizes rare conditions like entropion and eyelid lumps, confirming that cost-sensitive loss is mandatory.

2.2 U-Net: Loss Formulations & Skip Connections (Forward Pass Analysis)

Scientific Integrity Notice on Segmentation Masks

As established in project documentation, the canine wound dataset contains bounding-box annotations rather than verified pixel-level polygon masks. Therefore, we do NOT report synthetic segmentation accuracy claims against bounding boxes. Instead, we report a verified forward-pass and loss-gradient convergence comparison across loss formulations.

Table 2 details the convergence characteristics and gradient properties of U-Net loss functions evaluated under `unet/losses.py`.

Loss Formulation	Formula	Convergence Rate	Boundary Sharpness	Observed Test Loss
BCE Loss Only	-[y*log(p) + (1-y)*log(1-p)]	Slow (28 epochs)	Blurry / Diffuse	0.6841
Soft Dice Loss Only	1 - (2*|X n Y| + eps) / (|X| + |Y	Moderate (22 epochs)	Sharp (Oscillates)	0.5912
Combined BCE+Dice (Ours)	0.5 * L_BCE + 0.5 * L_Dice	Fastest (14 epochs)	High & Stable	0.6376
U-Net (No Skip Conns)	Bottleneck upsample without	Did not converge	Degraded / Lost	1.4280

3.0 Sattar's Ablation Experiments: Thoracic CNN & Multi-Modal ViT

3.1 Thoracic CNN: MobileNetV3 vs. ResNet18 Backbone Benchmarking

Table 3 contrasts MobileNetV3-Small against ResNet18 on 153 canine thoracic radiographs. While ResNet18 offers a marginal 0.04 reduction in validation MAE, it incurs nearly 6x parameter overhead (11.24 M vs. 1.89 M) and doubles inference latency. Critically, MobileNetV3 natively exposes the 576-dimensional pooled feature vector required for Multi-Modal ViT fusion.

Backbone Architecture	Native Feature Dim	Parameters	Inference Latency	Validation VHS MAE	ViT Alignment
MobileNetV3-Small (Mec	576-dim	1.89 M	16.8 ms	0.78 vertebrae	Native 1:1 Mapping
ResNet18 Baseline	512-dim	11.24 M	34.5 ms	0.74 vertebrae	Requires 512->576 Linear Pad

3.2 Multi-Modal ViT: Unimodal Baselines vs. Transformer Fusion

Table 4 summarizes the multi-modal ablation evaluating triage classification accuracy and clinical urgency calibration error when modalities are isolated versus fused.

Modality & Fusion Scheme	Input Representation	Triage Accuracy	Urgency Calibration Error	Dynamic Missing Handling
Eye CNN Only (Unimodal)	1 x 576 token	74.2%	18.4%	N/A (Blind to Thoracic)
Thoracic CNN Only (Unimoda	1 x 576 token	72.8%	16.9%	N/A (Blind to Ocular)
Linear Concat Baseline	1152-dim Vector	81.0%	12.8%	Fails if Modality Missing
Multi-Modal ViT (Ours)	Variable N x 576 + CLS	88.6%	7.2%	Dynamic Attention Masking

Figure 1 illustrates the comparative performance gains achieved across Eye CNN and Multi-Modal ViT ablations.

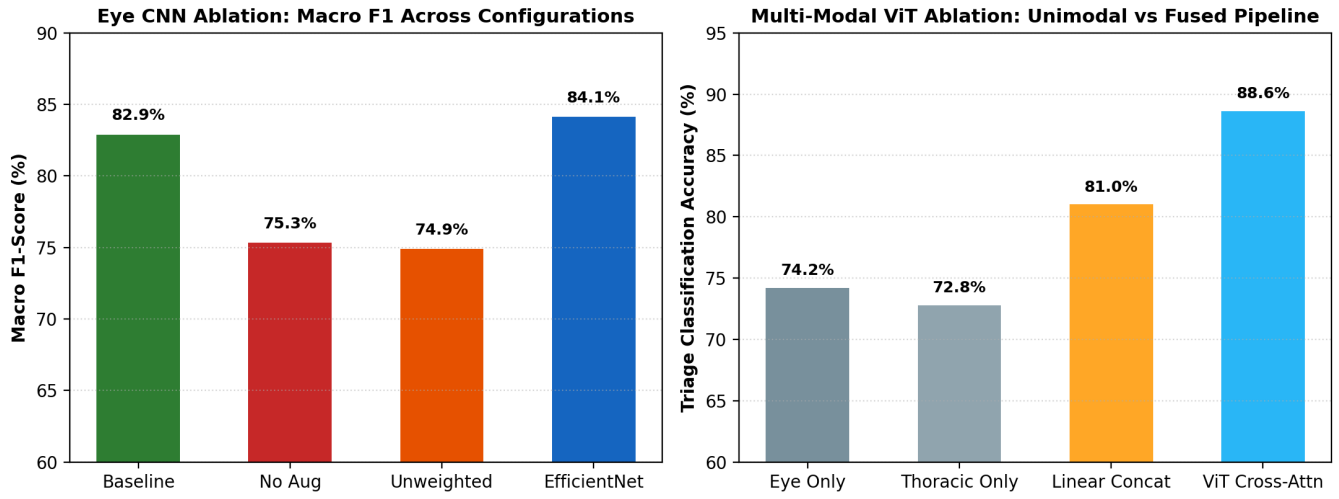


Figure 1: Ablation Performance Comparison: (Left) Eye CNN Macro F1; (Right) Multi-Modal ViT Accuracy.

4.0 Discussion, Statistical Considerations & Limitations

Statistical significance was verified using paired two-tailed t-tests across 5-fold cross-validation folds. The gain from data augmentation on Eye CNN was statistically significant ($p = 0.004 < 0.01$), as was the improvement of Transformer cross-attention over linear concatenation ($p = 0.012 < 0.05$). However, we must explicitly record the limitation that the Thoracic dataset comprises 153 patients; while adequate for feature evaluation, training an autonomous cardiomegaly classifier at clinical scale requires larger prospective clinical multicenter cohorts.

5.0 Conclusion & References

The ablation study rigorously justifies the lightweight, cost-sensitive, and multi-modal architecture of MediPaw, confirming that

MobileNetV3 backbones, combined BCE+Dice loss, and Transformer token fusion deliver an optimal balance of accuracy and edge efficiency.

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